An Internship Report

Of

Full Stack Internship — ETS Project

Submitted

To

Diverse Loopers

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)

By

Roll Number of Student: 2410031117

Name of Student: NIYATI MITTAL

Batch: 2024–2028

Candidate's Declaration

I, Niyati Mittal, do hereby declare that the internship report titled "Full Stack Internship — ETS Project" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: Niyati Mittal

Roll No.: 2410031117

Acknowledgement

I am using this opportunity to express my gratitude to everyone who supported me throughout the internship. I am thankful for their aspiring guidance, invaluably constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the team at Diverse Loopers, and in particular Ashish Kumar Tiwari (CMGO), for providing me the opportunity and the mentorship to gain hands-on exposure in full-stack development across the ETS and Crash Course projects. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-

Student Name: Niyati Mittal

Roll No.: 2410031117

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3. Internship Completion Certificate

DIVERSE LOOPERS

CERTIFICATE OF INTERNSHIP

This certificate acknowledges your outstanding contribution and dedication during internship, showcasing your commitment to excellence, innovation, and teamwork.

This is proudly presented to

NIYATI MITTAL

on completion of internship for the role of **FULL STACK INTERN** from **30April 2025 to 30 July 2025.**

Presented on
01 Aug 2026

Ashish Kumar Tiwari
CMGO

4. Project Description

4.1 Introduction

This report presents a detailed account of the Full Stack Internship undertaken at Diverse Loopers, in the domain of "ETS — Exception Tracking System / Recruiter Verified Workforce Identity." The internship was carried out from 30 April 2025 to 30 July 2025 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The primary aim of this internship was to build a strong practical foundation in full-stack web development, and to apply it to real-world platform-engineering problems such as internal exception-tracking dashboards, workflow automation, and public-facing product landing pages.

The internship followed a project-based structure combining frontend UI design, backend/data integration, and iterative feedback from the team, culminating in a functioning internal dashboard and a set of shipped UI improvements. On successful completion of the internship, a Certificate of Internship was issued by Diverse Loopers.

4.2 Organization Profile

Diverse Loopers is the organization at which this internship was undertaken. The team builds internal platform tools and recruiter-facing products, including the ETS (Exception Tracking System / Recruiter Verified Workforce Identity) product — an internal system for tracking, reviewing, and resolving data and workflow exceptions — alongside educational/content initiatives such as the Crash Course project. The internship was supervised under the guidance of Ashish Kumar Tiwari, CMGO.

4.3 Problem Statement

Internal teams handling exceptions arising from automated workflows (duplicate identities, database write failures, document upload failures, claim failures, and automation failures) previously lacked a single, transparent view of system health and case status. Reviewers had no consolidated way to search, filter, assign, and resolve exceptions, and duplicate-identity signals were not visible alongside the exception records they related to. The problem addressed by this internship was to design and build an internal dashboard that consolidates exception data, exposes system-health metrics at a glance, and streamlines the review-and-resolution workflow — while, in parallel, improving the accessibility and responsiveness of the Crash Course learning content delivered on the web.

4.4 Project Objectives

- To design and develop the ETS Dashboard/Overview page with key system-health metrics.
- To implement metric cards for Total Exceptions, Open, Under Review, and Resolved cases.
- To build a visual exception-type breakdown for quick system-health insight.
- To design and develop the Exceptions Management page with search and filtering.
- To implement inline reviewer assignment, automation retry triggers, and a resolution workflow with notes.
- To integrate duplicate-identity data into the exceptions view for faster, more informed resolution.
- To integrate AI-generated summaries for key actions (assign, retry, resolve).

- To review and improve the email/notification schema for clearer, actionable alerts.
- To convert subject-wise PDF notes into structured, responsive HTML pages for the Crash Course project.
- To ensure UI consistency and responsiveness across devices for both projects.

4.5 Scope of the Project

The scope of this internship spans the ETS internal dashboard end-to-end — from the Overview page and system-health metrics, through Exceptions Management with search, filtering, assignment and resolution workflows, to the integration of duplicate-identity data and AI-generated action summaries. It further covers a review of the notification/email schema used by the system. In parallel, the scope includes UI and content work on the Crash Course project: poster layout design, conversion of PDF notes into responsive HTML pages, and responsive-design fixes across the interface. The internship was limited to internal-tool development and did not involve production deployment decisions or infrastructure ownership.

4.6 Technologies and Tools Used

- Frontend: React.js, Next.js, JavaScript, HTML5, CSS3, Tailwind CSS
- Backend: Node.js, REST APIs
- Data: PostgreSQL, SQL
- AI Integration: AI/LLM APIs (action summaries)
- Tooling: Git & GitHub

4.7 System Architecture

The general architecture followed during the internship for the ETS product can be summarized in the following layers:

- Data Layer: PostgreSQL tables — including exceptions and duplicate_identity — store exception records, statuses, and duplicate-identity signals.
- Integration Layer: duplicate_identity fields (match reason, identity fingerprint, detection date) are merged into DUPLICATE_IDENTITY exception records.
- Workflow Layer: Reviewer assignment, automation retry triggers, and a resolution workflow with notes operate on exception records and update their status (OPEN, UNDER_REVIEW, RESOLVED).
- Intelligence Layer: AI/LLM APIs generate contextual summaries for key actions such as assign, retry, and resolve.
- Presentation Layer: A React/Next.js dashboard renders the Overview page, metric cards, exception-type breakdown, and the Exceptions Management page with search and filters.
- Notification Layer: Email schema and templates alert relevant stakeholders on exception and resolution events.

4.8 Methodology

The internship followed a project-based methodology, moving iteratively through design, implementation, and review across the two projects, as summarized in the table below.

Project	**Focus**
Phase 1 — ETS Dashboard/Overview	Designed and built the Dashboard/Overview page, Hero Section, and system-health metric cards (Total Exceptions, Open, Under Review, Resolved).
Phase 2 — Exceptions Management	Built the Exceptions Management page: search, filters by exception type/status, inline reviewer assignment, automation retry triggers, and a resolution workflow with notes.
Phase 3 — Data Integration	Integrated duplicate-identity data from the duplicate_identity table into the exceptions view — match reason, identity fingerprint, and detection date.
Phase 4 — AI-Assisted Actions	Integrated AI-generated summaries for key actions (assign, retry, resolve) to give internal teams clearer context.
Phase 5 — Notifications Review	Reviewed the existing email schema and notification templates, and proposed improvements to data fields and message content.
Phase 6 — Crash Course Content	Converted subject-wise PDF notes into structured, responsive HTML pages and designed poster layouts for the project.
Phase 7 — UI Polish	Worked on UI fixes, improvements, and responsive design across screen sizes and devices for the Crash Course interface.

4.9 Expected Outcomes

- Ability to design and build an internal, data-driven dashboard from metrics to detailed record views.
- Practical understanding of exception-management workflows, including assignment, automation retries, and resolution.
- Hands-on exposure to integrating related datasets (duplicate-identity records) into a primary workflow view.
- Experience integrating AI/LLM-generated summaries into an operational tool to aid team decision-making.
- Capability to review and improve notification/email schema for clarity and actionability.
- Skill in converting static content (PDF notes) into structured, responsive web pages.
- A completed, in-production-style internal dashboard and a set of shipped UI/responsiveness improvements, resulting in a Certificate of Internship from Diverse Loopers.

4.10 Certificates of Completion and Communication Proof

The Certificate of Internship issued by Diverse Loopers, confirming successful completion of the Full Stack Internship from 30 April 2025 to 30 July 2025, is attached in Chapter 3 of this report.

5. Bibliography/References

1. Diverse Loopers, "Certificate of Internship — Full Stack Intern," issued to Niyati Mittal, presented 01 August 2026.

2. React Documentation, https://react.dev/

3. Next.js Documentation, https://nextjs.org/docs

4. Tailwind CSS Documentation, https://tailwindcss.com/docs

5. PostgreSQL Documentation, https://www.postgresql.org/docs/

6. Node.js Documentation, https://nodejs.org/en/docs